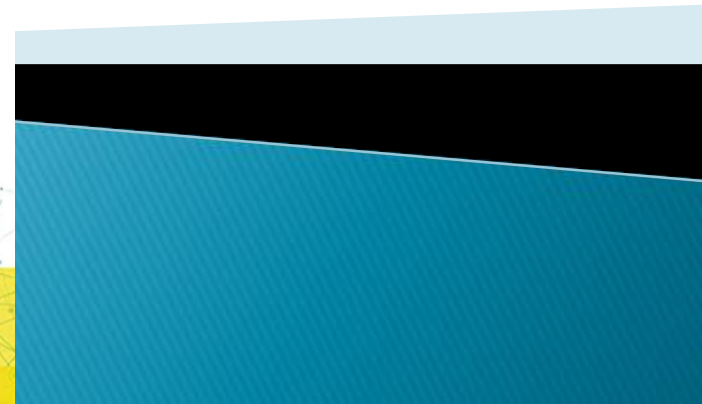
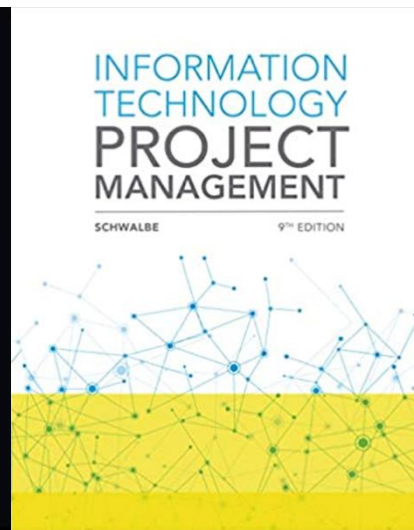


Chapter 1: Introduction to Project Management

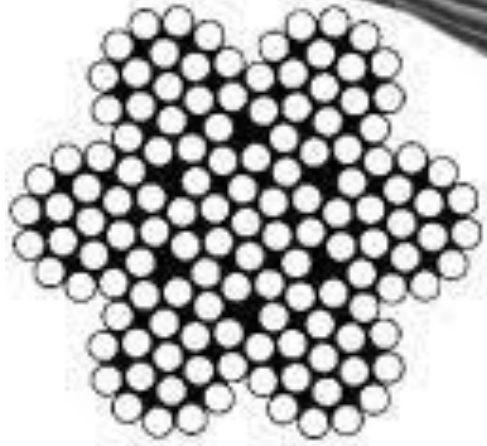
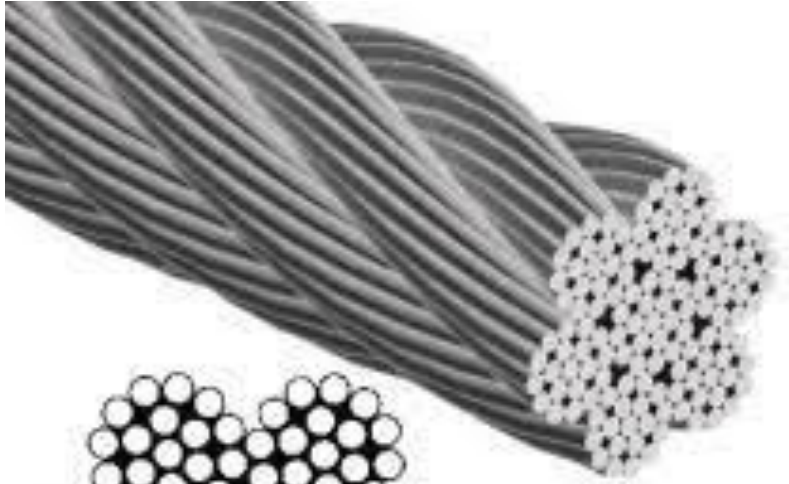
Information Technology Project Management



Introduction

- ▶ Many organizations today have a new or renewed interest in project management
- ▶ Computer hardware, software, networks, and the use of **interdisciplinary and global work teams** have radically changed the work environment
- ▶ The world as a whole spends **nearly 25% of its \$126.3 trillion gross product on projects of all kinds**
- ▶ More than **40 million people (2026)** regard project management as their profession

Engineering Vs. Project Management



Project Management Statistics

- ▶ The overall information and communications technology market grew by 6 percent to almost **\$3 trillion** in 2010
- ▶ In the U.S. the size of the **IT workforce topped 4 million** workers in 2008 (**and 10 million in 2026**), and the **unemployment** rate for IT professionals is half the rate for the overall labor market
- ▶ In 2011 the total compensation for the average senior project manager in U.S. dollars was \$105,000 per year in the United States (**\$195,000 per year + bonus in 2026**)
- ▶ The number of people earning their Project Management Professional (PMP) certification continues to increase. **44 percent of employers listed project management as a skill** they looked for in new college grads, behind only communication and technical skills

Motivation for Studying Information Technology (IT) Project Management

- ▶ IT Projects have a terrible track record, as described in the What Went Wrong?
- ▶ A 1995 Standish Group study (CHAOS) found that **only 16.2% of IT projects were successful** in meeting scope, time, and cost goals; **over 31% of IT projects were canceled** before completion
- ▶ A PricewaterhouseCoopers study found that overall **half of all projects fail** and only **2.5% of corporations consistently meet their targets** for scope, time, and cost goals for all types of project.

Advantages of Using Formal Project Management

- ▶ Better control of financial, physical, and human resources
- ▶ Improved customer relations
- ▶ Shorter development times
- ▶ Lower costs
- ▶ Higher quality and increased reliability
- ▶ Higher profit margins
- ▶ Improved productivity
- ▶ Better internal coordination
- ▶ Higher worker morale

What Is a Project?

- ▶ A **project** is “a **temporary** endeavor undertaken to create a **unique** product, service, or result” (PMBOK® Guide, Fifth Edition, 2012)
- ▶ **Operations** is work done **continuously** to sustain the business
- ▶ Projects end when their objectives have been reached or the project has been terminated
- ▶ Projects can be large or small and take a short or long time to complete

Examples of IT Projects

- ▶ A team of students creates a smartphone application and sells it online
- ▶ A company develops a driverless car
- ▶ A small software development team adds a new feature to an internal software application for the finance department
- ▶ A college upgrades its technology infrastructure to provide wireless Internet access across the whole campus

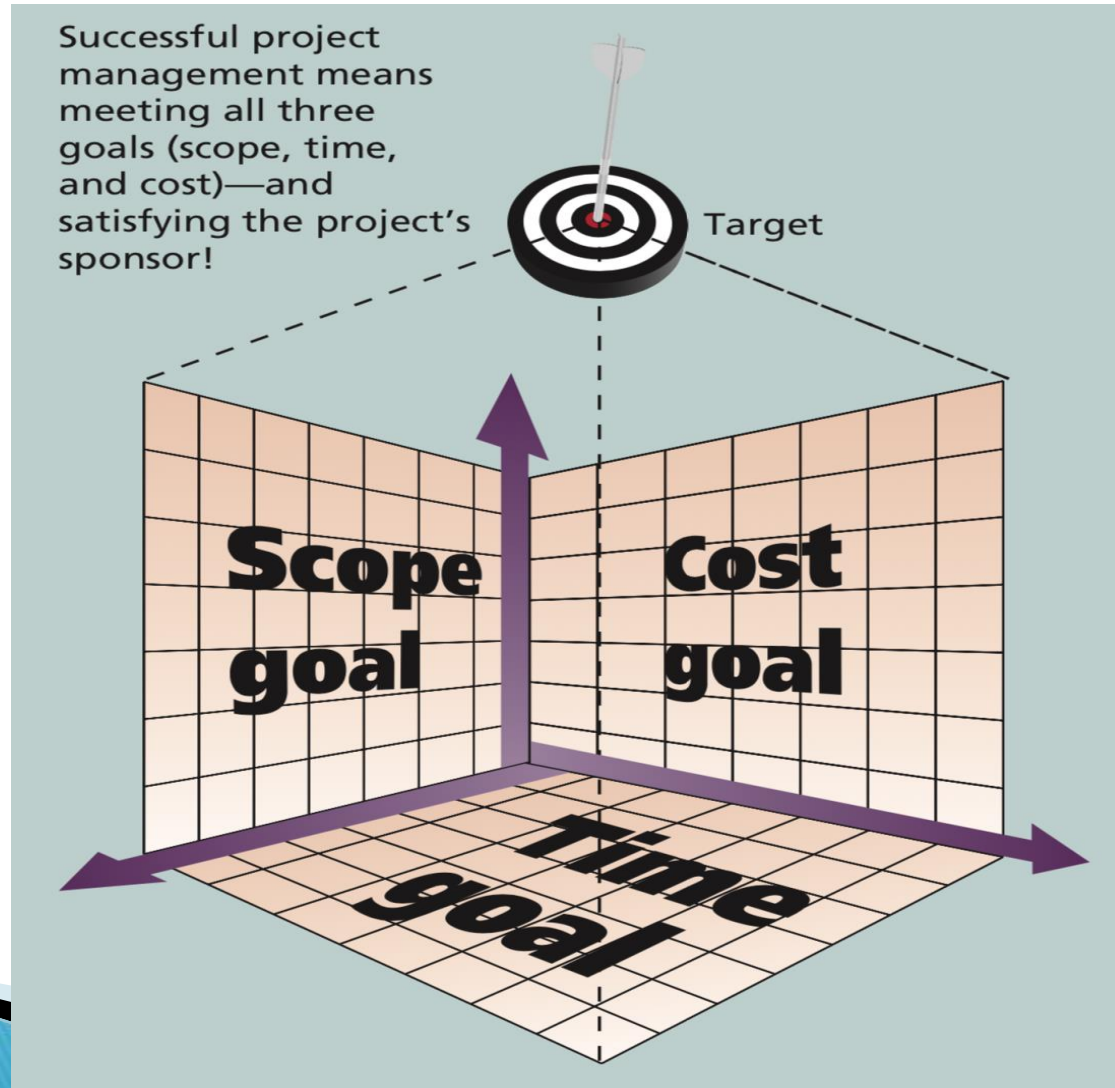
Project Attributes

- ▶ A project
 - has a unique purpose
 - is temporary, with clear timeline
 - is developed using **progressive elaboration**
 - requires resources, often from various areas
 - should have a primary customer or **sponsor**
 - The **project sponsor** usually provides the direction and funding for the project
 - involves uncertainty

Project and Program Managers

- ▶ **Project managers** work with project sponsors, project team, and other people involved in a project to meet project goals
- ▶ **Program:** group of related projects managed in a coordinated way to obtain benefits and control not available from managing them individually (PMBOK® Guide, Fifth Edition, 2012)
- ▶ **Program managers** oversee programs; often act as bosses for project managers

Figure 1-1 The Triple Constraint of Project Management



What is Project Management?

- ▶ **Project management** is “the application of **knowledge, skills, tools** and **techniques** to project activities to meet project requirements” (PMBOK® Guide, Fourth Edition, 2012)
- ▶ Project managers strive to meet the **triple constraint** (project scope, time, and cost goals) and also facilitate the entire process to meet the needs and expectations of project stakeholders

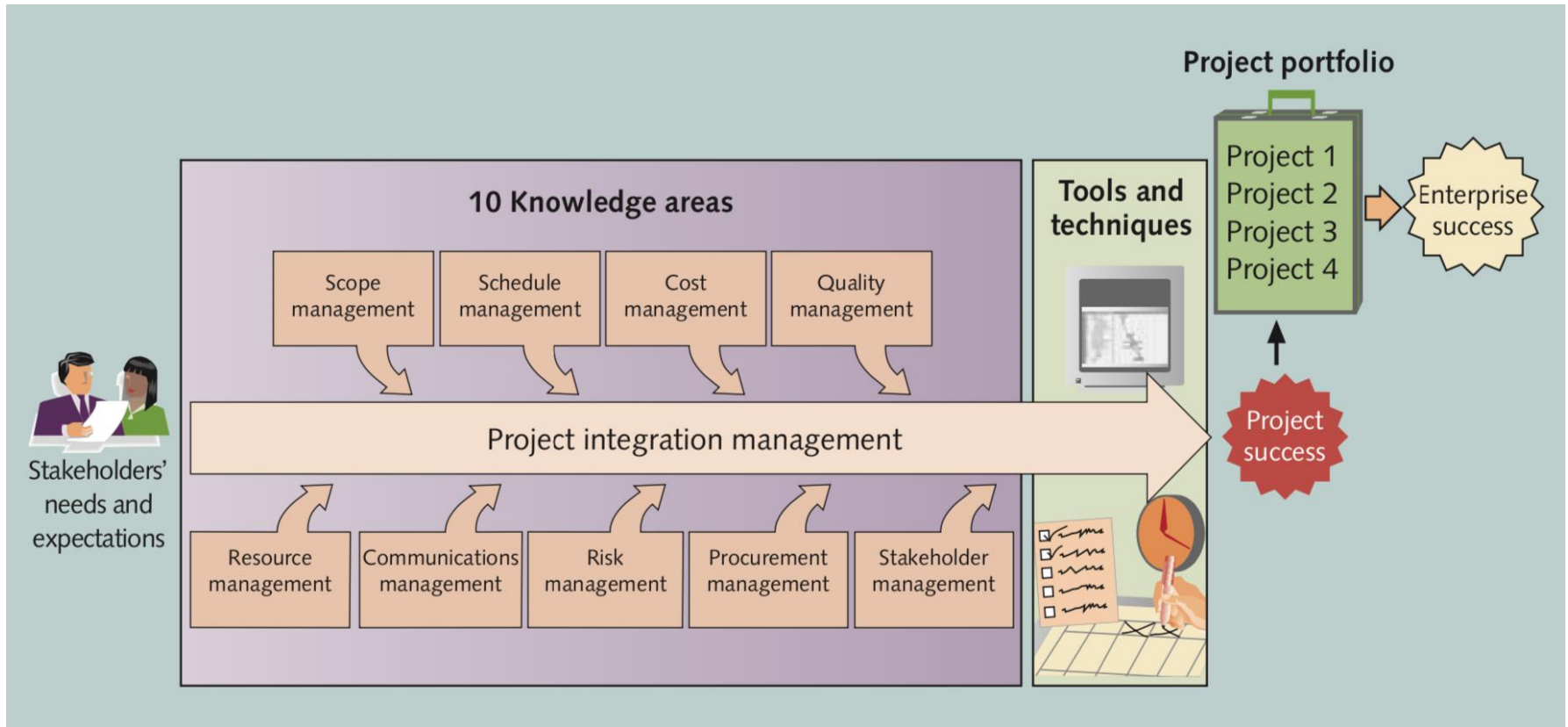
10 Project Management Knowledge Areas

- ▶ **Knowledge areas** describe the key competencies that project managers must develop
- ▶ Project managers must have knowledge and skills in all 10 knowledge areas (project integration, scope, time, cost, quality, human resource, communications, risk, procurement, and stakeholder management)
- ▶ This text includes an entire chapter on each knowledge area

Project Stakeholders

- ▶ **Stakeholders** are the people involved in or affected by project activities
- ▶ Stakeholders include
 - the project sponsor
 - the project manager
 - the project management team
 - the project team
 - support staff
 - customers
 - users
 - suppliers
 - opponents to the project

Figure 1-2 Project Management Framework



PROCESS GROUP AND KNOWLEDGE AREA MAPPING

Knowledge Areas	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	
9. Project Resource Management		9.1 Plan Resource Management 9.2 Estimate Activity Resources	9.3 Acquire Resources 9.4 Develop Team 9.5 Manage Team	9.6 Control Resources	
10. Project Communications Management		10.1 Plan Communications Management	10.2 Manage Communications	10.3 Monitor Communications	
11. Project Risk Management		11.1 Plan Risk Management 11.2 Identify Risks 11.3 Perform Qualitative Risk Analysis 11.4 Perform Quantitative Risk Analysis 11.5 Plan Risk Responses	11.6 Implement Risk Responses	11.7 Monitor Risks	
12. Project Procurement Management		12.1 Plan Procurement Management	12.2 Conduct Procurements	12.3 Control Procurements	
13. Project Stakeholder Management	13.1 Identify Stakeholders	13.2 Plan Stakeholder Engagement	13.3 Manage Stakeholder Engagement	13.4 Monitor Stakeholder Engagement	

Knowledge Areas	Project Management Process Groups				
	Initiating Process Group	Planning Process Group	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work	4.4 Monitor and Control Project Work 4.5 Perform Integrated Change Control	4.6 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Time Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Resources 6.5 Estimate Activity Durations 6.6 Develop Schedule		6.7 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Perform Quality Assurance	8.3 Control Quality	
9. Project Human Resource Management		9.1 Plan Human Resource Management	9.2 Acquire Project Team 9.3 Develop Project Team 9.4 Manage Project Team		
10. Project Communications Management		10.1 Plan Communications Management	10.2 Manage Communications	10.3 Control Communications	
11. Project Risk Management		11.1 Plan Risk Management 11.2 Identify Risks 11.3 Perform Qualitative Risk Analysis 11.4 Perform Quantitative Risk Analysis 11.5 Plan Risk Responses		11.6 Control Risks	
12. Project Procurement Management		12.1 Plan Procurement Management	12.2 Conduct Procurements	12.3 Control Procurements	12.4 Close Procurements
13. Project Stakeholder Management	13.1 Identify Stakeholders	13.2 Plan Stakeholder Management	13.3 Manage Stakeholder Engagement	13.4 Control Stakeholder Engagement	

▶ https://www.researchgate.net/figure/Project-Management-Process-group-and-Knowledge-Area-Mapping-PMI-2013_fig1_325574821

Project Management Tools and Techniques

- ▶ **Project management tools and techniques** assist project managers and their teams in various aspects of project management
- ▶ Specific tools and techniques:
 - Project charter, scope statement, and WBS (scope)
 - Gantt charts, network diagrams, critical path analysis, critical chain scheduling (time)
 - Cost estimates and earned value management (cost)
 - See Textbook for many more

Super Tools

- ▶ **“Super tools”** are those tools that have high use and high potential for improving project success, such as:
 - Software for task **scheduling** (such as project management software)
 - **Scope** statements
 - **Requirements** analyses
 - **Lessons-learned** reports
- ▶ Tools already extensively used that have been found to improve project importance include:
 - Progress reports
 - Kick-off meetings
 - Gantt charts
 - Change requests

What Went Right? Improved Project Performance

The Standish Group's CHAOS studies show improvements in IT projects in the past decade:

- ▶ The number of **successful** IT projects has more than doubled, from 16 percent in 1994 to 37 percent in 2010
- ▶ The number of **failed** projects decreased from 31 percent in 1994 to 21 percent in 2010
- ▶ Success rates were the highest ever in the most recent CHAOS study

Why the Improvements?

The reasons for the increase in successful projects vary.

the average **cost** of a project has been more than cut in half.

Better **tools** have been created to monitor and control progress

Better skilled project managers

Better management processes are being used.

The fact that there are processes is significant in itself.”*

Project Success

- ▶ There are several ways to define project success:
 - The project met scope, time, and cost goals
 - The project satisfied the customer/sponsor
 - The results of the project met its main objective, such as making or saving a certain amount of money, providing a good return on investment, or simply making the sponsors happy
 - Today: Add sustainability ...etc.

Table 1-2: What Helps Projects Succeed?*

1. User involvement
2. Executive support
3. Clear business objectives
4. Emotional maturity
5. Optimizing scope
6. Agile process
7. Project management expertise
8. Skilled resources
9. Execution
10. Tools and infrastructure

*The Standish Group, “CHAOS Activity News” (August 2011).

What the Winners Do...

- ▶ Recent research findings show that companies that excel in project delivery capability:
 - Use an **integrated project management toolbox** (use standard/advanced PM tools, lots of templates)
 - **Grow** project leaders, emphasizing business and **soft skills**
 - Develop a **streamlined** project delivery process
 - Measure **project health** using metrics, like customer satisfaction or return on investment

Program and Project Portfolio Management

- ▶ A **program** is “a group of related projects managed in a coordinated way to obtain benefits and control not available from managing them individually” (PMBOK® Guide, Fifth Edition, 2012)
- ▶ A **program manager** provides leadership and direction for the project managers heading the projects within the program
- ▶ Examples of common programs in the IT field include infrastructure, applications development, and user support

Suggested Skills for Project Managers

- ▶ The Project Management Body of Knowledge
- ▶ Application area knowledge, standards, and regulations
- ▶ Project environment knowledge
- ▶ General management knowledge and skills
- ▶ Soft skills or human relations skills

Table 1-3 Ten Most Important Skills and Competencies for Project Managers

1. People skills
2. Leadership
3. Listening
4. Integrity, ethical behavior, consistent
5. Strong at building trust
6. Verbal communication
7. Strong at building teams
8. Conflict resolution, conflict management
9. Critical thinking, problem solving
10. Understands, balances priorities

Importance of Leadership Skills

- ▶ Effective project managers provide leadership by example
- ▶ A **leader** focuses on long-term goals and big-picture objectives while inspiring people to reach those goals
- ▶ A **manager** deals with the day-to-day details of meeting specific goals
- ▶ Project managers often take on the role of both leader and manager

Figure 1-6. Sample Gantt Chart Created with Project 2010

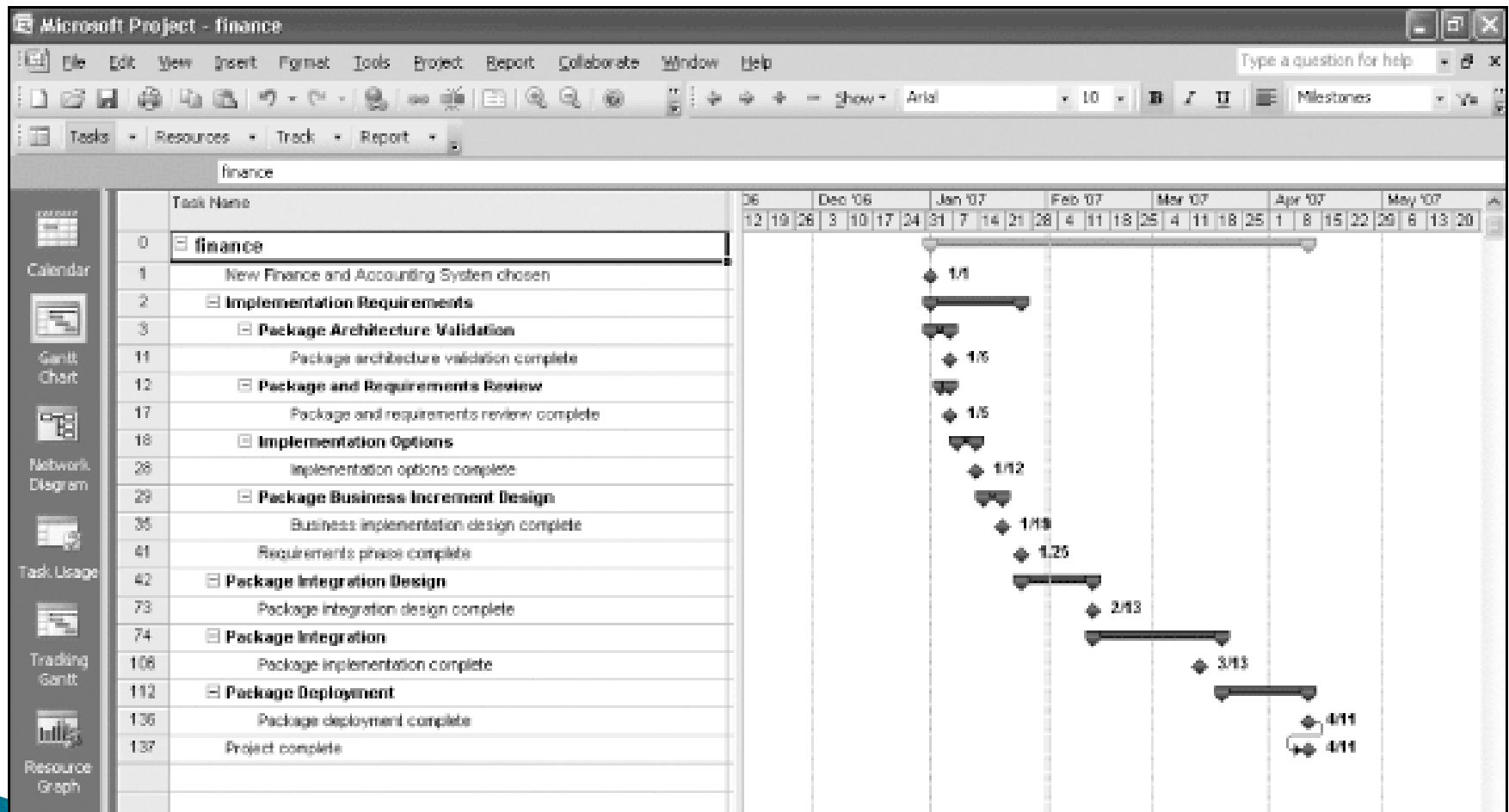
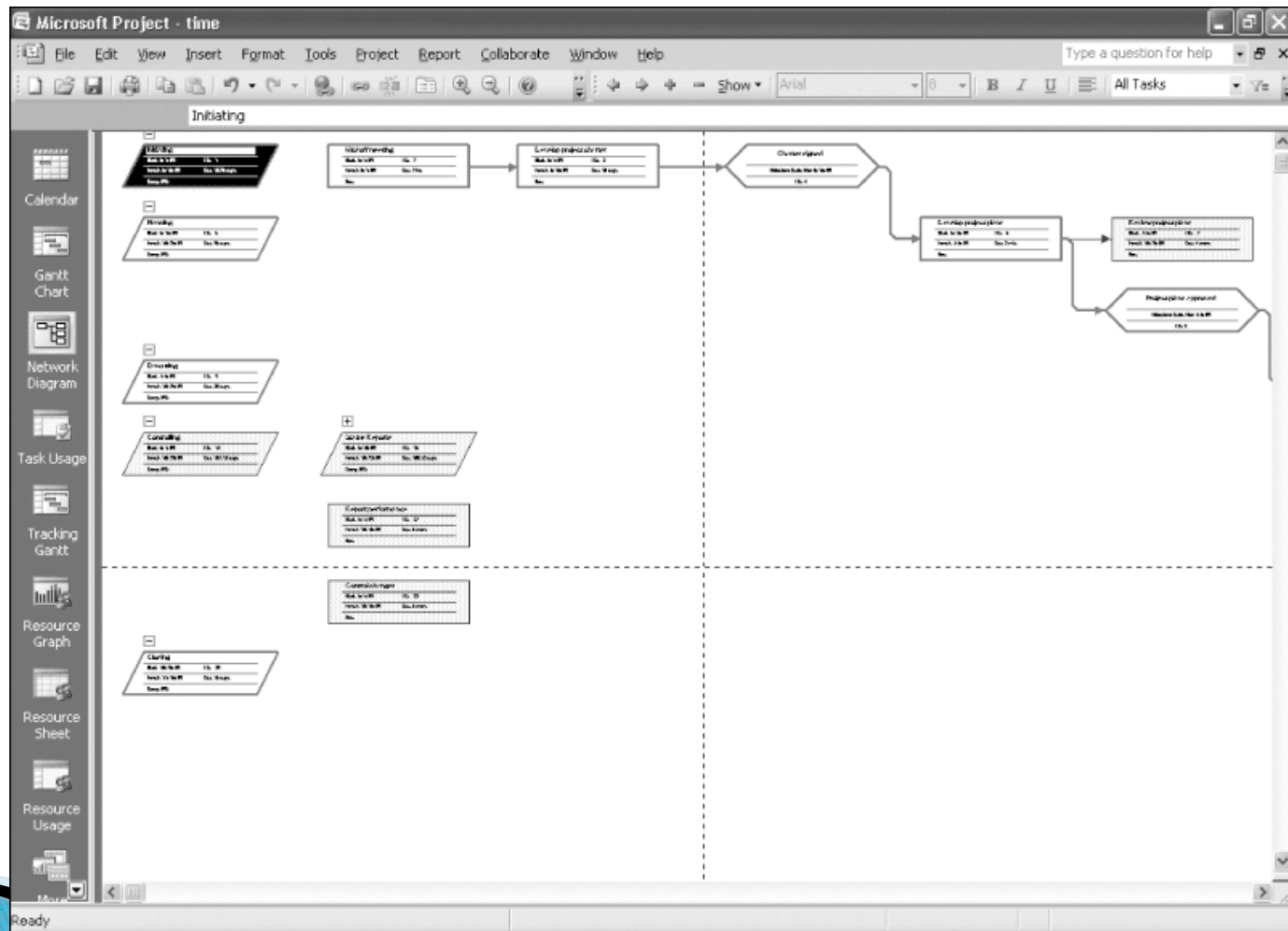


Figure 1-7. Sample Network Diagram Created with Project 2010



Project Management Offices

- ▶ In the 100s, many companies began creating PMOs to help them handle the increasing number and complexity of projects
- ▶ A **Project Management Office (PMO)** is an organizational group responsible for coordinating the project management function throughout an organization

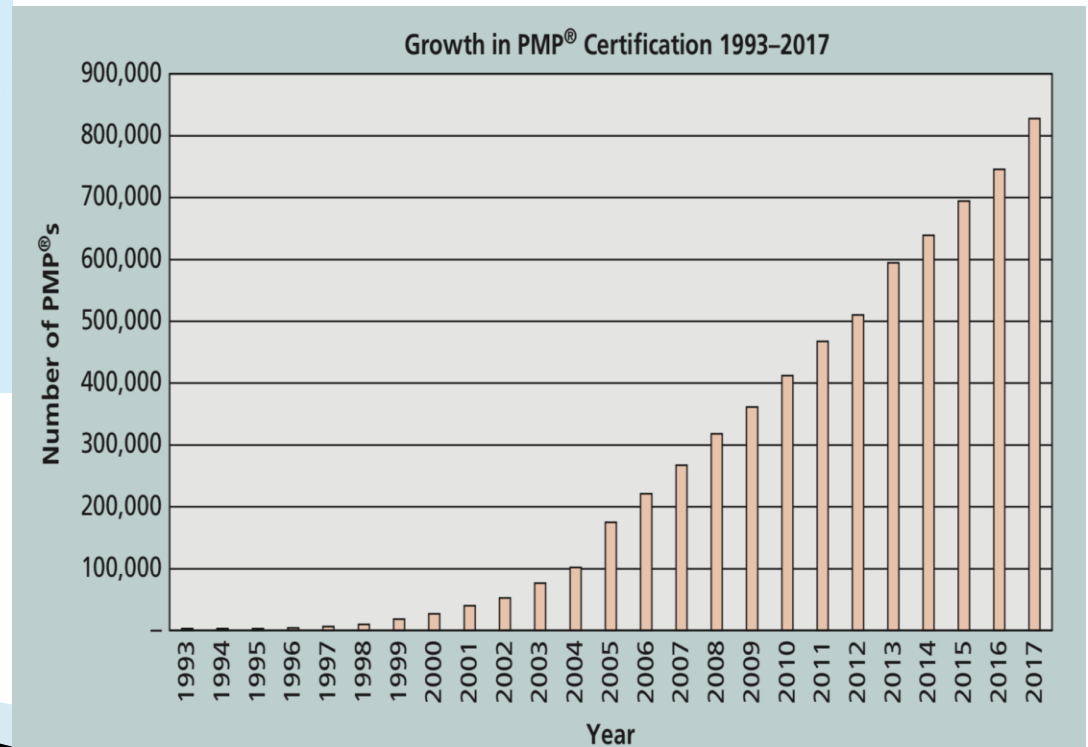
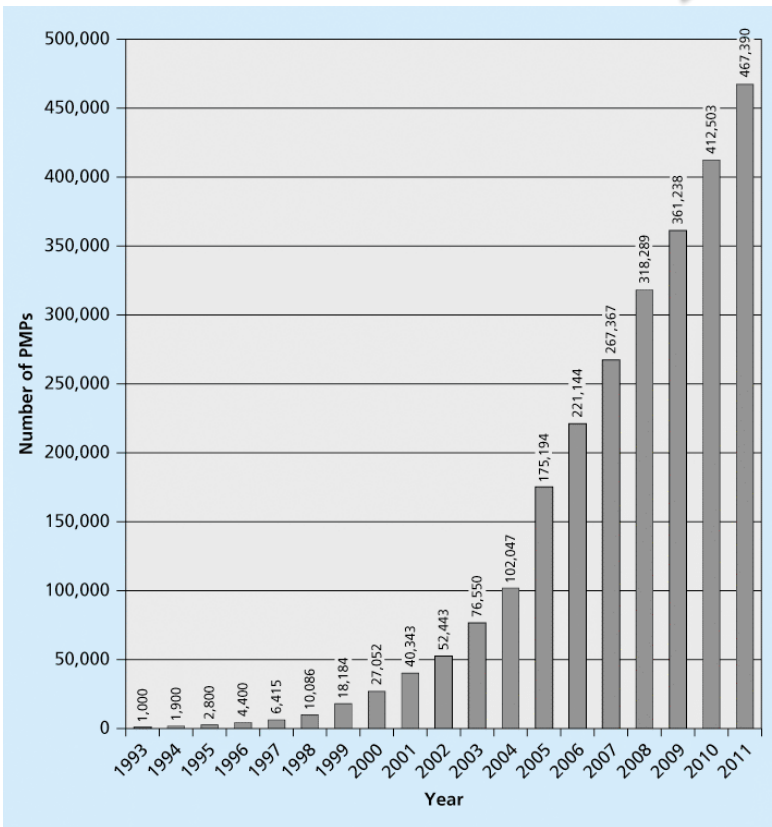
The Project Management Institute

- ▶ The Project Management Institute (PMI) is an international professional society for project managers founded in 1969
- ▶ PMI has continued to attract and retain members, reporting more than 380,000 members worldwide by 2012
- ▶ There are communities of practices in many areas, like information systems, financial services, and health care
- ▶ Project management research and certification programs continue to grow
- ▶ Students can join PMI at a reduced fee and earn the Certified Associate in Project Management (CAPM) certification(see www.pmi.org for details)

Project Management Certification

- ▶ PMI provides certification as a **Project Management Professional (PMP)**
- ▶ A PMP has documented sufficient project experience, agreed to follow a code of ethics, and passed the PMP exam
- ▶ The number of people earning PMP certification is increasing quickly
- ▶ PMI and other organizations offer additional certification programs (see Appendix B)

Figure 1-9 Growth in PMP Certification, 1993-2017



Ethics in Project Management

- ▶ **Ethics**, loosely defined, is a set of principles that guide our decision making based on personal values of what is “right” and “wrong”
- ▶ Project managers often face ethical dilemmas
- ▶ In order to earn PMP certification, applicants must agree to PMI’s Code of Ethics and Professional Conduct
- ▶ Several questions on the PMP exam are related to professional responsibility, including ethics